

2.4 Two Variable Relationships

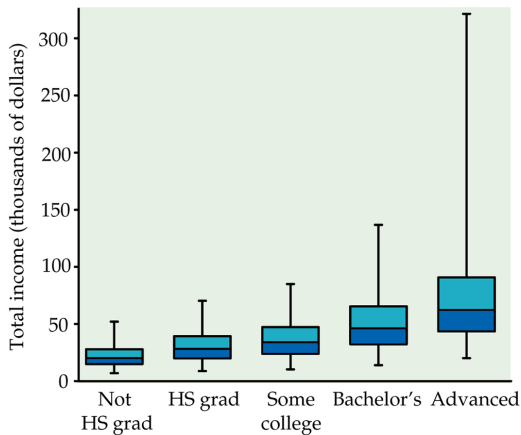
- Goal: We need to turn “raw” data into information.
- One qualitative (categorical) and one quantitative (numerical) variables
 - ① Visualization: side-by-side boxplot and histogram
 - ② Summary statistic: five-number summary for each category
- Two quantitative variables
 - ① Visualization: scatterplot
 - ② Summary statistic: correlation

One Qualitative & One Quantitative | Side-by-Side Comparison

- One is **qualitative (categorical)** and the other is **quantitative (numerical)**.
- The quantitative values can be viewed as **separate samples**, and each sample is identified by **each category** of the qualitative variable.
- Provide the summary statistics and graphics for side-by-side comparison.
 - 1 Statistics for comparison: measures of central tendency, measures of dispersion, five-number summary
 - 2 Graphs for comparison: dotplot, boxplot, histogram

Example | Side-by-Side Comparison

- Comparison of income (quantitative response variable) for different education levels (five categories).



Exercise | Side-by-Side Comparison

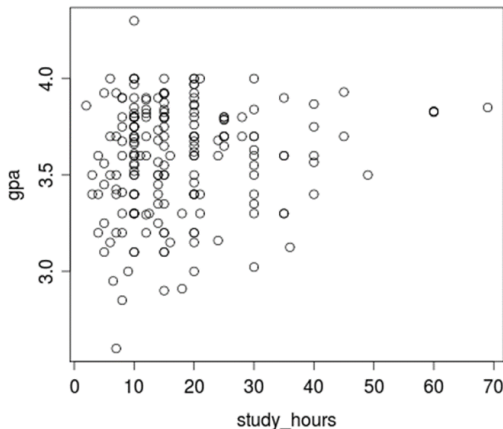
- A random sample of households from three different cities of northern Arizona was obtained and their electric bill for August was recorded. The data is given in the following table:

Kingman		Williams		Flagstaff	
23	39	34	36	38	45
31	40	34	36	40	46
33	41	35	37	42	50
37	42	35	39	43	58
38	50	36	41	44	65

- 1 Compare the distribution of electric bills for the 3 cities using boxplot.
- 2 The five-number summary for the three cities is
- 3 The mean and variance for these three cities are

Two Quantitative Variables | Scatterplot

- A **scatterplot** is a plot of pairs of quantitative values as points on a coordinate axis, showing the relationship between **two variables** measured on the **same cases**.



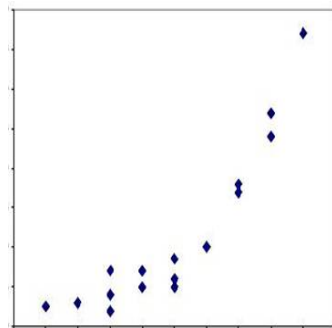
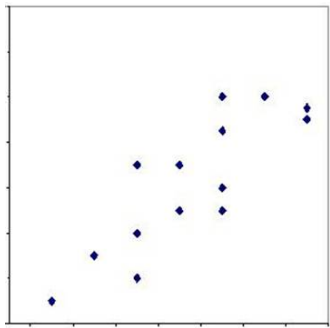
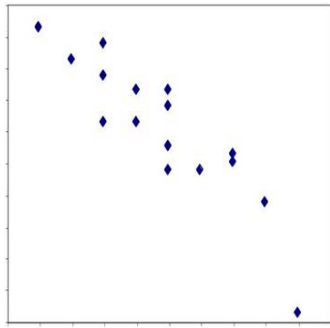
Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Scatterplot

- If the objective is to predict a response variable Y from an explanatory variable X , then by convention, we select Y as the vertical axis variable and X as the horizontal axis variable.
- Information from scatterplots
 - 1 **Type** of association: linear or nonlinear
 - 2 **Direction** of association: positive (direct) or negative (inverse)
 - 3 **Strength** of association: weak, moderate, or strong
 - 4 **Outliers** (individual values that fall outside the overall pattern of the relationship)

Type of Association

- Two variables have a **linear** relationship if the points in a scatterplot roughly follow a **straight** line with a **non-zero slope**.

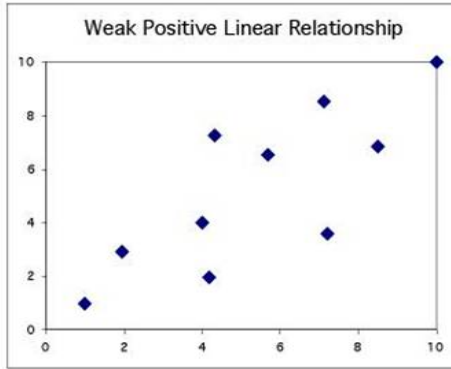
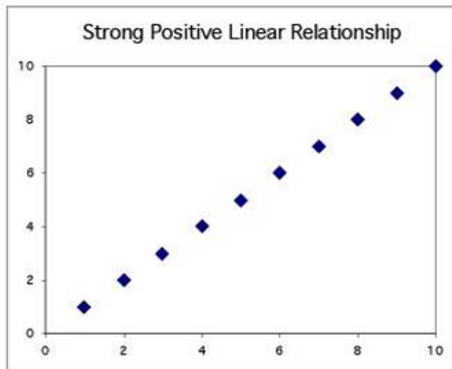


Direction of Association

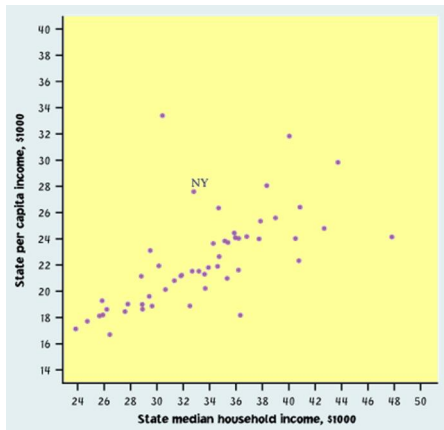
- Suppose **large** values of Y tend to have **large** values of X , and **small** values on X also tend to have **small** values on Y . This is a **positive association** or **direct relationship**.
 - Example: Height and weight – taller people tend to be heavier.
- Suppose **large** values of X tend to have **small** values of Y , and vice-versa. This is a **negative association** or **inverse relationship**.
 - Example: Smoking and life expectancy – people who smoke more cigarettes per day tend not to live as long as those who smoke fewer
- Two variables are **not associated** if knowing the value of one variable does not give you any information about the value of the other variable.
- Note: Statistical associations are overall tendencies.

Strength of Association

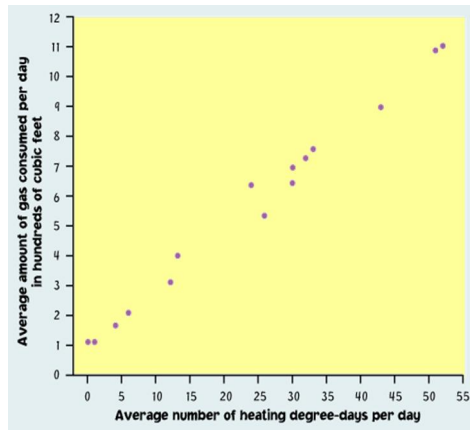
- **Strength** of a relationship is determined by how closely the points follow a clear form.



Example 2.13 | Strength of Association



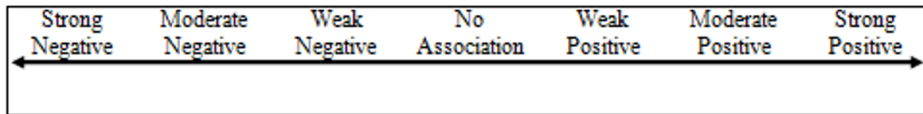
Note: This is a **weak** relationship. For a particular state median household income, you cannot predict the state per capita income very well.



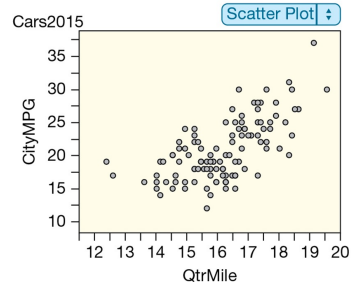
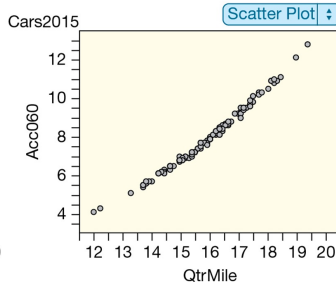
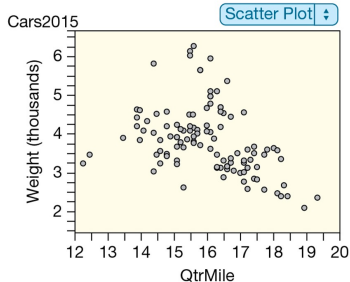
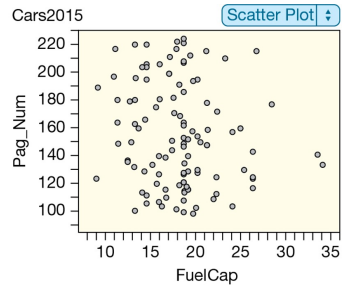
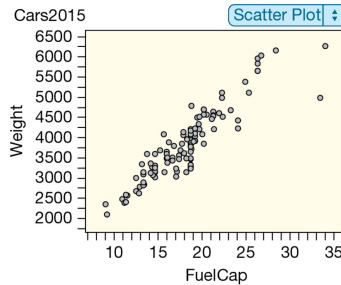
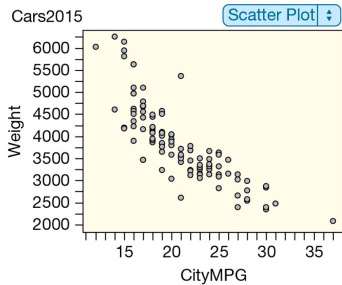
Note: This is a very **strong** relationship. The daily amount of gas consumed can be predicted quite accurately for a given temperature value.

Example 2.14 | Cars Data (Direction and Strength)

- Make initial guesses for the strength and direction of association for each of the following:
 - Weight (in pounds) vs. CityMPG
 - Weight (in pounds) vs. FuelCapacity (in gallons)
 - PageNum (page number in Consumer Reports) vs. FuelCapacity (in gallons)
 - Weight vs. QtrMile (time to go 1/4 mile in seconds)
 - Acc060 (acceleration time from 0 to 60 mph in seconds) vs. QtrMile
 - CityMPG vs. QtrMile

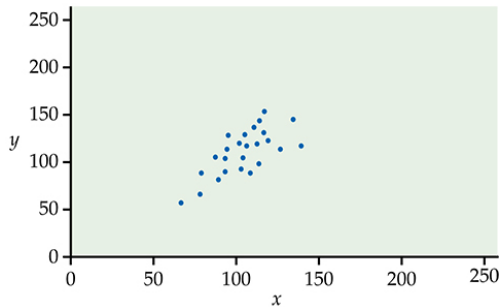
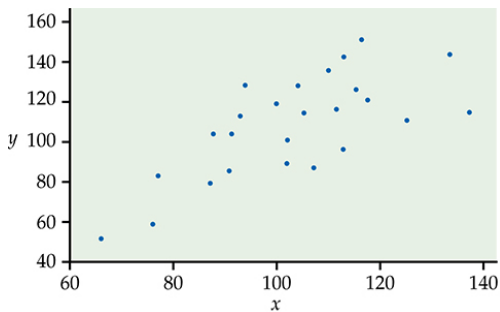


Example 2.14 | Cars Data (Direction and Strength)



Measure of Linear Relationship | Motivation

- Which one shows the stronger linear association? and why?



- Our eyes sometimes can be deceived. We need a [numerical measure of linear association](#) for quantifying the closeness of the scatter to a linear relation.

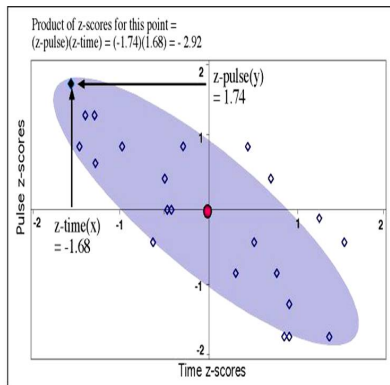
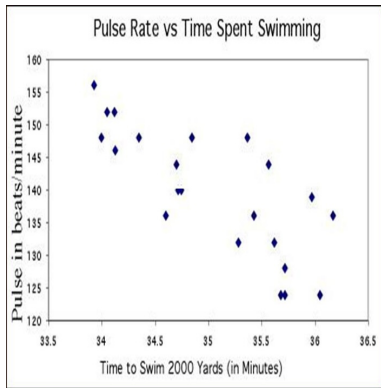
Measure of Linear Relationship | Correlation

- The **correlation** quantifies the **direction** and **strength** of the **linear relation** between **two numerical (quantitative) variables**.
- Sample correlation vs. Population correlation
 - 1 **Sample correlation** r
 - 2 **Population correlation** ρ ("rho")

Sample Correlation | Understanding

- r is an average of the products of the standardized values of X and Y :

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$



Sample Correlation | Computation

■ Computation of the sample correlation r :

- 1 There are data on variables X and Y from a sample of n subjects, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- 2 Obtain s_x and s_y , the sample standard deviations for X and Y variables:

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}, \quad s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}}.$$

- 3 Compute the **standardized observations** of x_i and y_i as follows:

$$\frac{x_i - \bar{x}}{s_x}, \quad \frac{y_i - \bar{y}}{s_y}.$$

- 4 Finally,

$$r = \frac{1}{n - 1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

Example 2.15 | Correlation computation

- We have the following $n = 5$ pairs of observations

$$(2, 5), (1, 3), (5, 6), (0, 2), (2, 4).$$

First, make a scatterplot of this bivariate data.

Example 2.15 (continued) | Correlation computation

- Compute the sample correlation r by following its definition.

S1: Use the following worksheet to compute $\bar{x}$, $\bar{y}$, s_x , and s_y .

	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	2	5	0	1	0	1
	1	3	-1	-1	1	1
	5	6	3	2	9	4
	0	2	-2	-2	4	4
	2	4	0	0	0	0
Total	10	20	0	0	14	10

S2: We compute

$$\bar{x} = \frac{\sum x_i}{n} = \frac{10}{5} = 2, \quad \bar{y} = \frac{\sum y_i}{n} = \frac{20}{5} = 4$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{14}{5 - 1}} = \sqrt{3.5} = 1.87$$

$$s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}} = \sqrt{\frac{10}{5 - 1}} = \sqrt{2.5} = 1.58$$

Example 2.15 (continued) | Correlation computation

■ (Continued)

S3: The following worksheet is made for the computation of r :

x	y	$\frac{x-\bar{x}}{s_x}$	$\frac{y-\bar{y}}{s_y}$	$\left(\frac{x-\bar{x}}{s_x}\right) \cdot \left(\frac{y-\bar{y}}{s_y}\right)$
2	5	0.000	0.632	0.000
1	3	-0.535	-0.632	0.338
5	6	1.604	1.266	2.031
0	2	-1.069	-1.266	1.353
2	4	0.000	0.000	0.000
Total		0	0	3.722

S4: Therefore the sample correlation r is

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{1}{5-1} \times 3.722 = \frac{3.722}{4} = 0.93$$

- **Interpretation:** $r = 0.93$ is positive and large. This implies a strong association where both X and Y tend to be small or both tend to be large.

Sample Correlation | Property 1

■ The **sign** of r indicates the **direction** of a linear relation:

- 1 $r > 0$ if x and y have a **positive** association (**direct** relationship).
- 2 $r < 0$ if x and y have a **negative** association (**inverse** relationship).
- 3 Why? Recall

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Sample Correlation | Property 2

- The **magnitude** of r indicates the **strength** of a linear relation: a large value of $|r|$, a value close to 1 or -1, represents a **strong** linear relation.
 - 1 $r = 1$ if all pairs of values (x, y) lie exactly on a **straight** line with a **positive** slope.
 - 2 $r = -1$ if all pairs of values (x, y) lie exactly on a **straight** line with a **negative** slope.
 - 3 $r = 0$ if there is no general linear tendency for y values to either increase or decrease with x values.

Sample Correlation | Property 3

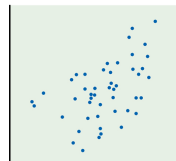
■ $-1 \leq r \leq 1$



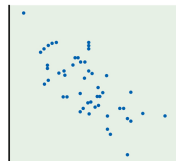
Correlation $r = 0$



Correlation $r = -0.3$



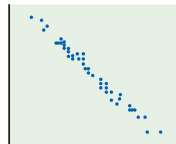
Correlation $r = 0.5$



Correlation $r = -0.7$

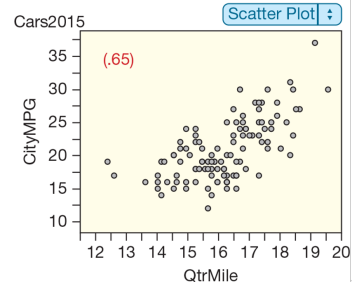
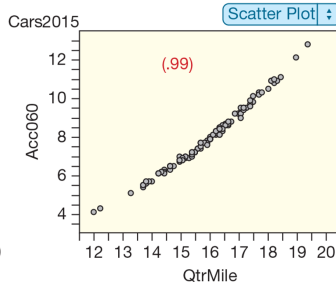
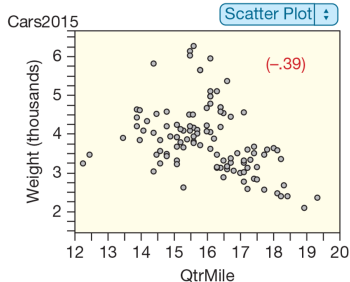
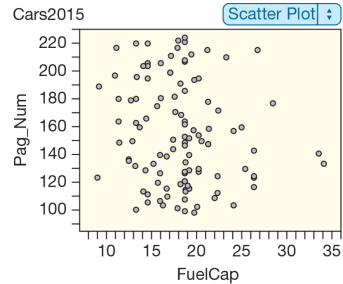
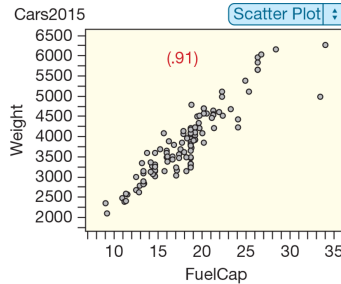
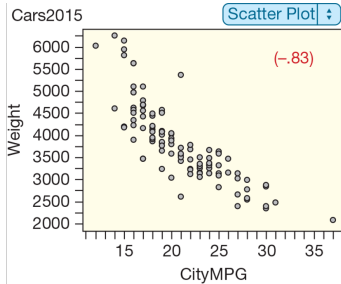


Correlation $r = 0.9$



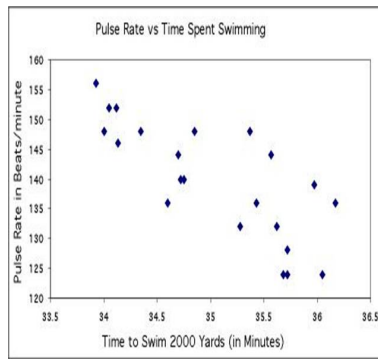
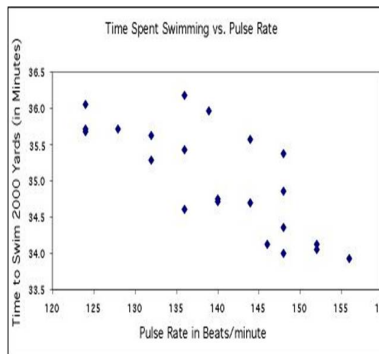
Correlation $r = -0.99$

Example 2.14 (revisited) | Cars Correlations



Sample Correlation | Property 4

- r does **not** distinguish X and Y .

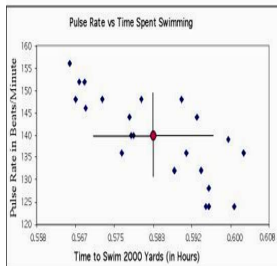
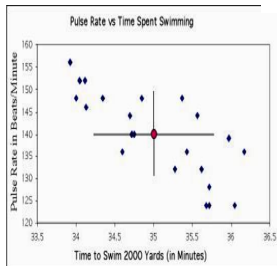


Note. “Time to swim” is the explanatory variable here, and belongs on the x-axis. However, in either plot r is the same ($r = -0.75$).

Sample Correlation | Property 5

■ r is unitless.

- 1 Changing the units of X and Y variables does **not** change the correlation coefficient r , because we get rid of all our units when we standardize (get z-scores).
- 2 Example. “Time to swim” is scaled on different units. However, in either plot r is the same ($r = -0.75$).

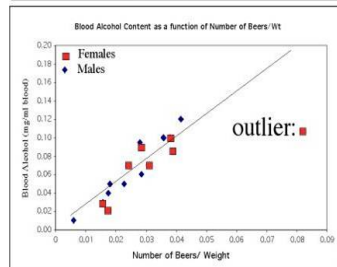
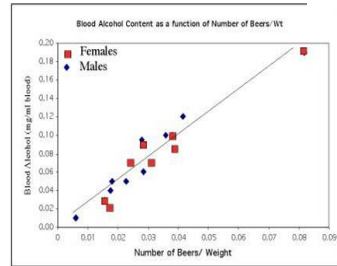


Sample Correlation | Property 6

- No matter how strong the association, r does **not** describe curved relationships.

Outliers in a Scatterplot

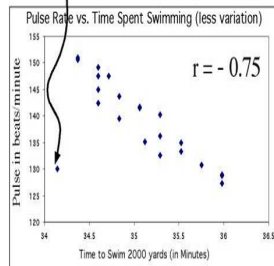
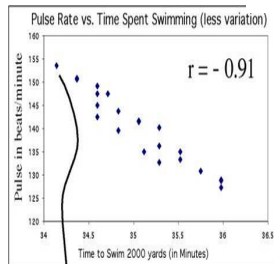
- The upper right-hand point here is **not** an outlier of the relationship. It is what you would expect for these many beers given the linear relationship between beers/weight and blood alcohol.
- The mid right-hand point is **not** in line with the others, so it **is** an outlier of the relationship.



Sample Correlation | Property 7

- Like the sample mean and sample standard deviation, the sample correlation is **not resistant**: r is strongly *affected* by a few *outliers*.

Note: Just moving one point away from the general trend here decreases the correlation from -0.91 (top) to -0.75 (bottom).

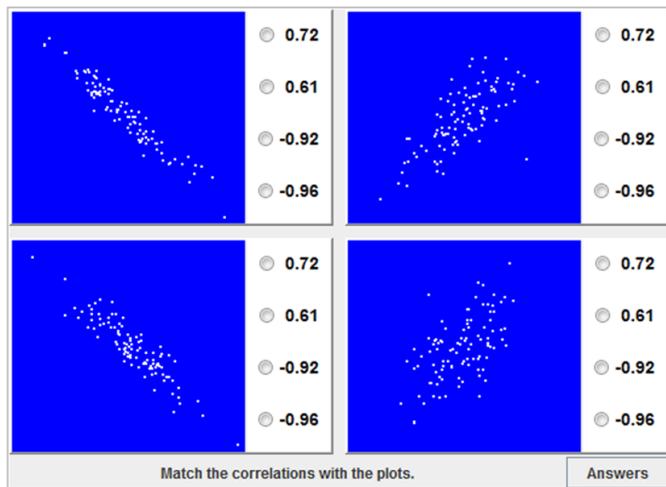


Correlation | Summary of Properties

- $-1 \leq r \leq 1$.
- The sign of r indicates the direction of association.
 - 1 Positive association: $r > 0$
 - 2 Negative association: $r < 0$
 - 3 No linear association: $r \approx 0$
- The closer r is to 1 or -1 , the stronger the linear association.
- r has no units and does not depend on the units of measurement.
- The correlation between X and Y is the same as the correlation between Y and X .
- r can be sensitive to outliers.
- Correlation does not imply causation!

Exercise 2.8 | Correlation Guessing Game

■ <https://www.guessthecorrelation.com/>



Source: Lock⁵ (2021), John Wiley & Sons, Inc.

Categorical Variable in Scatterplot



Temporary page!

$\text{\LaTeX}$ was unable to guess the total number of pages correctly. As there was some unprocessed material that should have been added to the final page this extra page has been added to receive it. If you rerun the document (without altering it) this surplus page will go away, because $\text{\LaTeX}$ now knows how many pages to expect for this document.